hydrology and topography

Purpose

This page documents the per-HRU hydrology and topography parameter files: `hydrology.hyd`, `topography.hyd`, and `field.fld`, plus the closely related shade-factor file `shade_factor.shf`. File names for the first three are set in `type input_hydrology` in `src/input_file_module.f90`; the shade file is set in `type shade_factor` in the same module.

Records in `hydrology.hyd`, `topography.hyd`, and `field.fld` are referenced by name from hru-data.hru (columns `hydro`, `topo`, and the field-name slot). Every HRU points at one record from each. Records in `shade_factor.shf` are used by the channel temperature routine in `src/ch_temp.f90`.

Each file is optional. If the slot in `file.cio` is `null` or the file is missing, the reader allocates a zero-length array and skips on. All four readers skip two header lines, then read one record per parameter set.

Source

- File names: src/input_file_module.f90, `type input_hydrology`, `type shade_factor`
- Readers:
  - src/hydrol_read.f90 (`hydrology.hyd`)
  - src/topo_read.f90 (`topography.hyd`)
  - src/field_read.f90 (`field.fld`)
  - src/shade_factor_read.f90 (`shade_factor.shf`)
- Type definitions: src/hydrology_data_module.f90, src/topography_data_module.f90, src/hydrograph_module.f90

hydrology.hyd

Per-HRU hydrology parameter sets (lateral flow, canopy, evaporation compensation, percolation, enrichment ratios). Read into `hyd_db` (`type hydrology_db` in `src/hydrology_data_module.f90`).

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“”	parameter set name
2	lat_ttime	real	days	0.0	exponential of the lateral flow travel time (0..120)
3	lat_sed	real	g/L	0.0	sediment concentration in lateral flow
4	canmx	real	mm H2O	0.0	maximum canopy storage
5	esco	real	none	0.0	soil evaporation compensation factor (0..1)
6	epco	real	none	0.0	plant water uptake compensation factor (0..1)
7	erorgn	real	none	0.0	organic N enrichment ratio (0 = model calculates per event)
8	erorgp	real	none	0.0	organic P enrichment ratio (0 = model calculates per event)
9	cn3_swf	real	none	0.0	soil water at CN3 (0 = field capacity, 0.99 = near saturation)
10	biomix	real	none	0.0	biological mixing efficiency, applied at end of each calendar year
11	perco	real	none	0.0	percolation coefficient, linear adjustment to daily perc (0..1)
12	lat_orgn	real	ppm	0.0	organic N concentration in lateral flow
13	lat_orgp	real	ppm	0.0	organic P concentration in lateral flow
14	pet_co	real	none	1.0	radiation coefficient used in Hargreaves equation
15	latq_co	real	none	0.3	lateral soil flow coefficient, linear adjustment to daily lateral flow (0..1)

Example from `refdata/Osu_1hru/hydrology.hyd`:

```
hydrology.hyd: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                 lat_ttime       lat_sed       can_max          esco          epco   orgn_enrich   org
hyd0001                0.00000       0.00000       1.00000       0.95000       0.50000       0.00000
```

Editor column names (`can_max`, `harg_pet`, `bio_mix`) differ from source field names. The reader uses position only.

topography.hyd

Per-HRU topography parameter sets. Read into `topo_db` (`type topography_db` in `src/topography_data_module.f90`).

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“default”	parameter set name
2	slope	real	m/m	0.02	average slope steepness in HRU
3	slope_len	real	m	50.0	average slope length for erosion
4	lat_len	real	m	50.0	slope length for lateral subsurface flow
5	dis_stream	real	m	100.0	average distance to stream
6	dep_co	real	none	1.0	deposition coefficient

Example from `refdata/Osu_1hru/topography.hyd`:

```
topography.hyd: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        slp       slp_len       lat_len      dist_cha         depos
topohru0001             0.01078      10.00000      10.00000     121.00000       0.00000
```

The same type and array are also used for landscape units; `hru-data.hru` and the subbasin objects both reference entries in `topo_db` by name.

field.fld

Per-field geometry used by the wind-erosion routine. Read into `field_db` (`type fields_db` in `src/topography_data_module.f9`

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“default”	parameter set name
2	length	real	m	500.0	field length for wind erosion
3	wid	real	m	100.0	field width for wind erosion
4	ang	real	deg	30.0	field angle for wind erosion

Example from `refdata/Osu_1hru/field.fld` (first four records shown):

```
field.fld: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                       len            wd           ang
fld011                500.00000     100.00000      30.00000
fld012                500.00000     100.00000      30.00000
fld031                500.00000     100.00000      30.00000
fld032                500.00000     100.00000      30.00000
```

The reader uses a `read .. backspace .. read` pattern, but the file format is one record per line, same as the other files on this page.

shade_factor.shf

Time- and landscape-unit-resolved shade factor used by the channel water-temperature routine. Read into `shf_db` (`type shade_factor_data` in `src/hydrograph_module.f90`). Activated per channel by `sf_on = 1` in `temperature.cha`; when `sf_on = 0` the channel uses the calibration-file value instead.

#	Column	Type	Units	Default	Description
1	jday	int	day of year	0	day of year the record applies to
2	lsu	int	none	0	landscape unit number
3	value	real	none	0.0	shade factor value

In `src/ch_temp.f90` the array is scanned for the matching `(jday, lsu)` pair and the `value` is assigned to `ssff`. There is no name column.

No example dataset in `refdata/` ships a `shade_factor.shf` file.

Related files

- `file.cio`. The `hru` line selects `hydrology.hyd`, `topography.hyd`, and `field.fld`; the shade file is listed separately.
- `hru-data.hru`. Every HRU references one entry from each of `hydrology.hyd`, `topography.hyd`, and `field.fld`.
- `temperature.cha`. Controls whether `shade_factor.shf` is consulted (`sf_on` flag).